



50 Python Practice Exercises

1. Basics & Syntax

1. Write a script that prints `"Hello, Python!"`.
 2. Assign a number to a variable and print its double using an f-string.
 3. Experiment with indentation: write a script that prints even numbers from 1–10.
 4. Run a Python script both in the **interactive shell** and from a `.py` file.
 5. Demonstrate the difference between single quotes `' '` and double quotes `" "` for strings.
-

2. Numbers & Types

6. Store an `int`, `float`, and `complex` number in variables and print their types.
 7. Try to add an `int` and a `str` → catch and explain the error.
 8. Convert a float `3.14` into an integer.
 9. Check if two different variables with the same value are identical using `is`.
 10. Test whether a variable is `None` using both `== None` and `is None`.
-

3. Data Structures

11. Create a list of 5 numbers and print the first, last, and a slice of the middle.
12. Reverse a list using slicing.
13. Modify an element inside a list and append a new one.
14. Create a tuple of 5 numbers → try modifying it and explain what happens.

15. Write a function that returns both quotient and remainder of division → unpack the result.
 16. Create a dictionary of 3 people with their ages → update one and add a new person.
 17. Check if a given key exists in a dictionary.
 18. Create a set of 5 numbers → add a duplicate and observe what happens.
 19. Use set `union()` and `intersection()` with another set.
 20. Count how many unique letters are in the word "mississippi" using a set.
-

4. Boolean Logic & Operators

21. Write a condition to check if a number is both positive and even.
 22. Compare `a == b` vs `a is b` with lists.
 23. Use `in` to check if "apple" exists in a list of fruits.
 24. Use `not in` to check if "grape" is missing from a set.
 25. Write a one-liner `if` statement that prints "yes" if a number is greater than 10.
-

5. Control Flow

26. Write an `if-elif-else` block that prints whether a number is **negative, zero, or positive**.
27. Loop through a list of 5 items and print them one by one.
28. Use `range(5)` to print numbers 0–4.
29. Use `range(start, end, step)` to print even numbers between 0 and 10.
30. Write a `while` loop that counts down from 5 to 1.
31. Write a `while` loop that stops when a random number equals 7.

32. Use `break` to exit a loop when encountering a negative number in a list.

33. Use `continue` to skip printing odd numbers in a loop.

6. Functions

34. Write a function that returns the square of a number.

35. Write a function with a default argument (`name="Python"`) that prints `"Hello, <name>".`

36. Write a function that accepts two numbers and returns both their sum and difference.

37. Write a function that accepts any number of arguments (`*args`) and prints them.

38. Write a function that accepts keyword arguments (`**kwargs`) and prints them.

39. Mix fixed arguments and `*args` in a single function.

40. Demonstrate keyword arguments by calling a function out of order.

7. Modules & Packages

41. Import the `datetime` module and print today's date.

42. From `math`, import `sqrt` and use it to compute $\sqrt{25}$.

43. Write your own module `mymodule.py` with a function `greet()`. Import it in another file.

44. Create a package with a submodule and call a function inside it.

45. Show what happens when you add a `print` statement at the top level of a module and import it.

8. List Comprehensions & Generators

46. Create a list of squares from 1–10 using list comprehension.
47. Filter only even numbers from 1–20 using list comprehension.
48. Use a generator expression to lazily produce numbers from 1–1,000,000 and get the first 5.
49. Write a generator function (`yield`) that produces Fibonacci numbers.
50. Combine list comprehension with an `if` condition to get words longer than 3 letters from a sentence.